

Kinematic Analysis of the KUKA LBR MED Surgical Manipulator

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1 Denavit–Hartenberg Convention

1.1 Robot Description

The KUKA LBR Med 7 R800 is a 7-DOF redundant surgical manipulator with all revolute joints. Its kinematic structure consists of alternating perpendicular joint axes, giving a 7-DOF configuration for a 6-DOF task space, with one degree of intrinsic redundancy.

Table 1: Link dimensions.

Parameter	Value	Description
d_1	0.340 m	Base platform to shoulder (joint 2 axis)
d_3	0.400 m	Shoulder to elbow (joint 4 axis)
d_5	0.400 m	Elbow to wrist center (joint 6 axis)
d_7	0.126 m	Wrist center to flange

1.2 Frame Assignment (Standard DH Convention)

Following the standard DH frame assignment procedure used in class:

1. Number joint axes $z_0, \dots, z_6$ where z_i is the axis of joint $i + 1$.
2. Place Frame 0 at the base with z_0 pointing vertically upward.
3. For $i = 1, \dots, 6$, place origin O_i at the intersection of z_{i-1} and z_i .
4. Define x_i along the common normal from z_{i-1} to z_i , directed from joint i to joint $i + 1$.
5. Define y_i to complete the right-handed frame.
6. Define Frame 7 at the end-effector with z_7 aligned with z_6 and O_7 at the flange center.

Key structural observation: All consecutive joint axes intersect, so all link offsets satisfy $a_i = 0$.

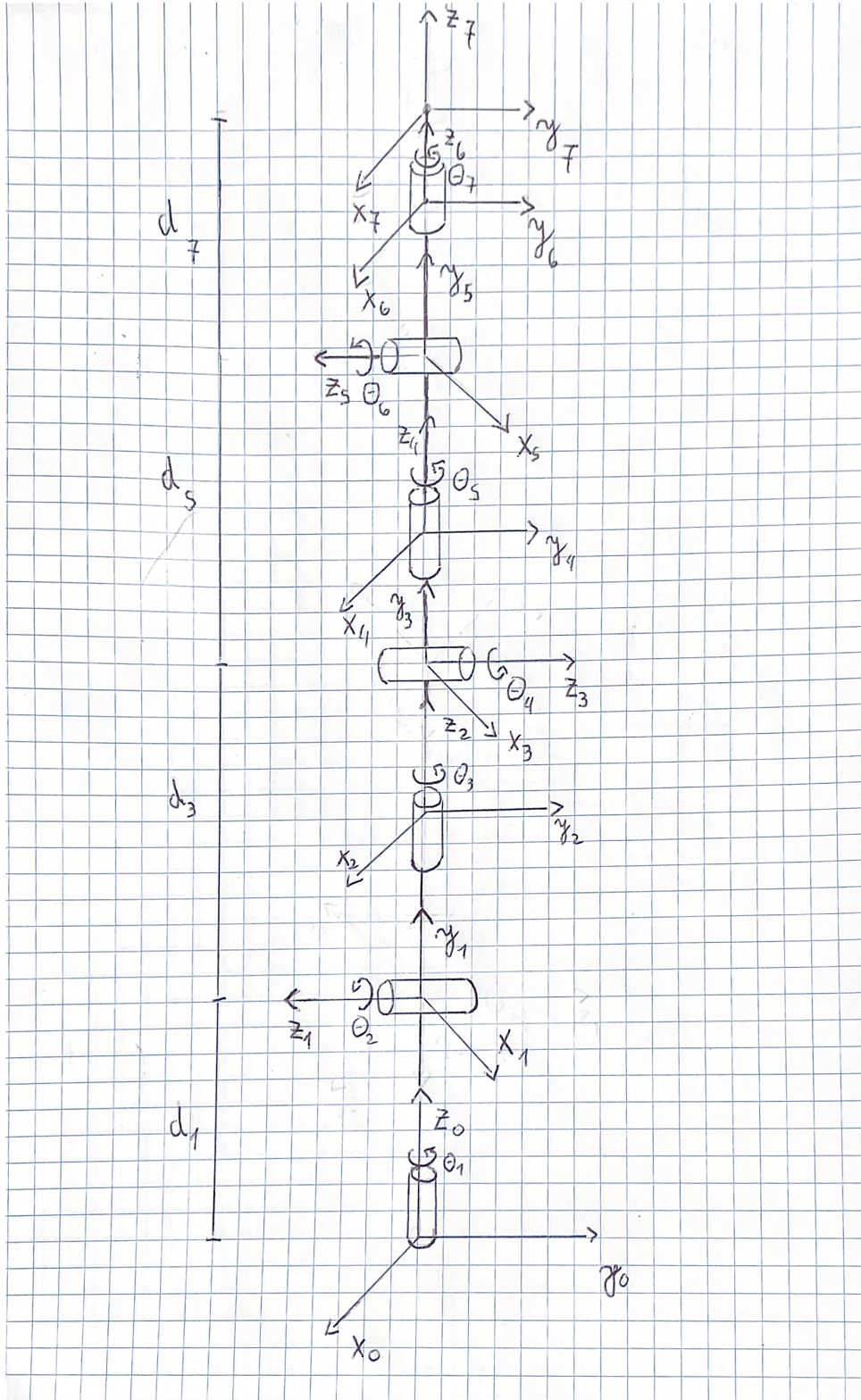


Figure 1: Hand-drawn DH frame assignment of the KUKA LBR MED.

1.3 DH Parameter Table

The DH transformation per link is:

$$A_i^{i-1}(q_i) = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i) \quad (1)$$

Table 2: Standard DH parameters for the KUKA LBR MED.

Link i	a_i (m)	α_i (rad)	d_i (m)	θ_i	Joint
1	0	$+\pi/2$	0.340	q_1	Base rotation
2	0	$-\pi/2$	0	q_2	Shoulder tilt
3	0	$-\pi/2$	0.400	q_3	Upper arm rotation
4	0	$+\pi/2$	0	q_4	Elbow tilt
5	0	$+\pi/2$	0.400	q_5	Forearm rotation
6	0	$-\pi/2$	0	q_6	Wrist tilt
7	0	0	0.126	q_7	End-effector rotation

2 Direct Kinematics

2.1 Method

Using the provided Symbolic Robotics MATLAB Toolbox, the direct kinematics transformation matrix is computed as:

$$T_7^0(q) = A_1^0(q_1)A_2^1(q_2) \cdots A_7^6(q_7) \quad (2)$$

2.2 Simulink Model

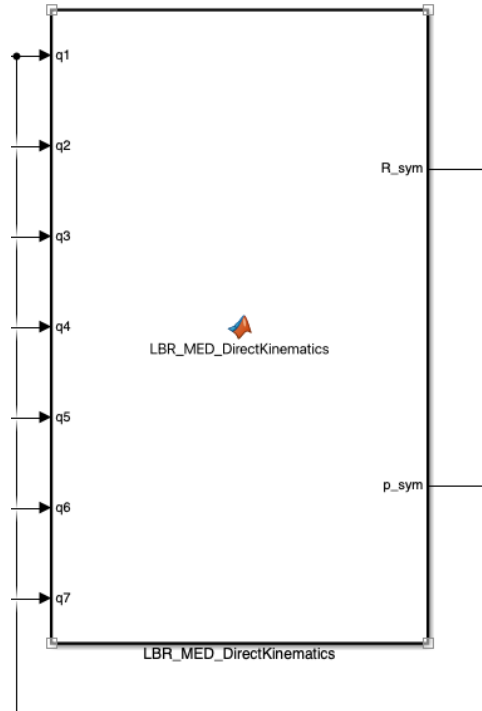


Figure 2: LBR_MED_DirectKinematics block in Simulink.

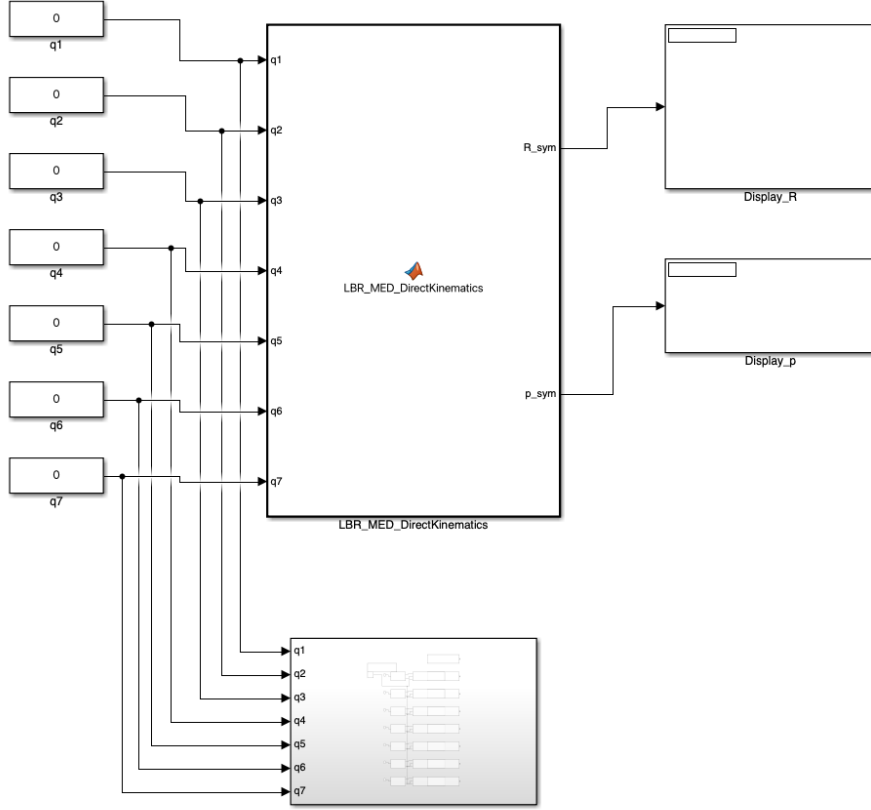


Figure 3: Complete validation model for Direct Kinematics (LBR_MED_Simul.slx).

2.3 Validation

Four configurations were chosen where the end-effector pose can be determined by geometric reasoning.

Table 3: Direct kinematics validation configurations.

Config	Joint angles q (rad)	Expected p (m)	Expected R	Error $\ p\ $
1: Home	all = 0	$[0, 0, 1.266]$	I_3	$< 10^{-12}$
2: Shoulder bend	$q_2 = \pi/2$, rest = 0	$[-0.926, 0, 0.340]$	$R_y(-\pi/2)$	$< 10^{-12}$
3: Base spin	$q_1 = \pi/2$, rest = 0	$[0, 0, 1.266]$	$R_z(\pi/2)$	$< 10^{-12}$
4: Elbow fold	$q_2 = q_4 = \pi/2$, rest = 0	$[-0.400, 0, 0.866]$	I_3	$< 10^{-12}$

3 Inverse Kinematics (Closed-Form)

3.1 Kinematic Decoupling Approach

The robot has a spherical wrist formed by joints 5, 6, and 7: axes z_4 , z_5 , and z_6 intersect at point O_6 . This enables kinematic decoupling:

- Arm problem (joints 1–4): position the wrist center O_6 .
- Wrist problem (joints 5–7): orient the end-effector.

Redundancy resolution: The robot has 7 DOF for a 6-DOF task. This report fixes $q_3 = 0$ to obtain a unique solution, reducing the arm problem to 3 unknowns (q_1, q_2, q_4).

3.2 Closed-Form Solution

Step 1 – Wrist center

The wrist center is the origin of frame 6, offset from the end-effector along the approach vector $a_e = R_d[:, 3]$:

$$p_W = p_e - d_7 a_e \quad (3)$$

Step 2 – Base rotation q_1

$$q_1 = \text{atan2}(p_{Wy}, p_{Wx}) \quad (\text{or } q_1 + \pi \text{ for the back-arm branch}) \quad (4)$$

Step 3 – Elbow angle q_4

Projecting the wrist center into the sagittal plane gives:

$$X = p_{Wx} \cos q_1 + p_{Wy} \sin q_1, \quad Z = p_{Wz} - d_1 \quad (5)$$

From the law of cosines applied to the triangle O_2 – O_4 – O_6 :

$$\cos q_4 = \frac{X^2 + Z^2 - d_3^2 - d_5^2}{2d_3d_5} \quad (6)$$

Two solutions arise from $\sin q_4 = \pm \sqrt{1 - \cos^2 q_4}$ (elbow-up and elbow-down).

Step 4 – Shoulder angle q_2

Expanding the sagittal-plane geometry yields the linear system

$$\begin{bmatrix} -(d_3 + d_5 c_4) & d_5 s_4 \\ d_5 s_4 & d_3 + d_5 c_4 \end{bmatrix} \begin{bmatrix} s_2 \\ c_2 \end{bmatrix} = \begin{bmatrix} X \\ Z \end{bmatrix} \quad (7)$$

which gives $q_2 = \text{atan2}(s_2, c_2)$ directly.

Steps 5–7 – Wrist angles q_5, q_6, q_7

With q_1 – q_4 known, the remaining rotation is isolated as:

$$R_{47} = (R_0^4)^\top R_d \quad (8)$$

where R_0^4 is the rotation part of $A_1 A_2 A_3 A_4$. Expanding the symbolic product $A_5 A_6 A_7$ (with $\alpha_5 = +\pi/2$, $\alpha_6 = -\pi/2$, $\alpha_7 = 0$) yields:

$$R_{47} = \begin{bmatrix} c_5 c_6 c_7 - s_5 s_7 & -c_5 c_6 s_7 - s_5 c_7 & -c_5 s_6 \\ s_5 c_6 c_7 + c_5 s_7 & -s_5 c_6 s_7 + c_5 c_7 & -s_5 s_6 \\ s_6 c_7 & -s_6 s_7 & c_6 \end{bmatrix} \quad (9)$$

Reading off the entries:

$$q_6 = \text{atan2}\left(\sqrt{1 - R_{47}(3, 3)^2}, R_{47}(3, 3)\right) \quad (10)$$

$$q_5 = \text{atan2}(-R_{47}(2, 3), -R_{47}(1, 3)) \quad (\text{when } \sin q_6 \neq 0) \quad (11)$$

$$q_7 = \text{atan2}(-R_{47}(3, 2), R_{47}(3, 1)) \quad (12)$$

A second wrist branch is obtained by negating $\sin q_6$ in (10) and flipping the signs in (11)–(12). Combined with the two arm branches (q_1 front/back) and two elbow branches (q_4 up/down), the solver generates up to **8 closed-form solutions**, returning the one closest to a reference configuration.

3.3 Singularity Conditions

Table 4: Closed-form IK singularity conditions.

Label	Condition	Geometric meaning
S1	$p_{Wx} = p_{Wy} = 0$	Wrist center on z_0 ; q_1 undefined.
S2	$ \cos q_4 = 1$	Arm fully extended or folded; q_2 undefined.
S3	$\sin q_6 = 0$	$z_4 \parallel z_6$; only $q_5 \pm q_7$ observable, not individually.

3.4 Simulink Model

The IK block (LBR_MED_IK.Simul.slx) takes the desired end-effector pose (p_d, R_d) as input, calls the analytical solver, and feeds the resulting joint vector through the direct kinematics block from Point 2 to verify round-trip consistency.

3.5 Validation

```
>> clear
>> LBR_MED_IK_validate
=== KUKA LBR Med 7 Analytical IK Validation ===

Testing Home position...
p_des: [0.0000, 0.0000, 1.2660]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]
q_ik:   [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]

Testing Random config (q3=0)...
p_des: [0.5110, 0.1086, 1.0467]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]
q_ik:   [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]

Testing Another random config (q3=0)...
p_des: [-0.5975, -0.6145, 0.4808]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]
q_ik:   [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]

All analytical IK validation tests PASSED.
>> |
```

Figure 4: Validation results for Inverse Kinematics showing errors below 10^{-8} .

Three test configurations (home, and two generic poses with $q_3 = 0$) were passed through $FK \rightarrow IK \rightarrow FK$. In all cases the round-trip position error $\|p_{IK} - p_d\|$ and orientation error $\|R_{IK} - R_d\|_F$ were below 10^{-8} , confirming correctness of the solver.

4 Geometric Jacobian

4.1 Definition

For each revolute joint i , the geometric Jacobian columns are:

$$J_{P_i} = z_{i-1} \times (p_e - O_{i-1}), \quad J_{O_i} = z_{i-1} \quad (13)$$

This relates joint velocities to end-effector velocity: $v_e = J(q)\dot{q}$.

4.2 Simulink Model

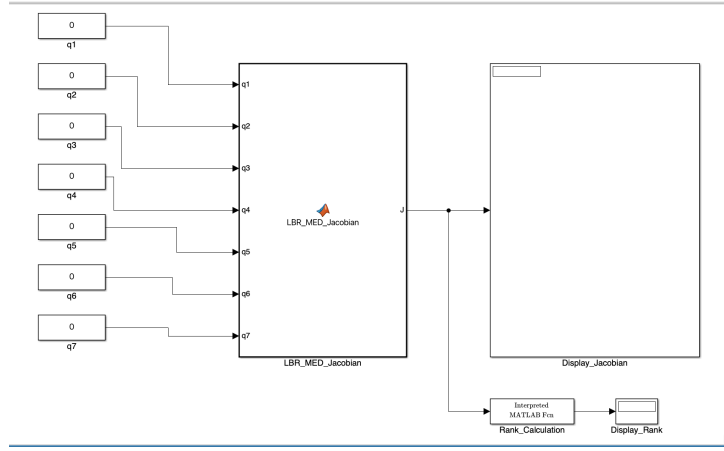


Figure 5: Jacobian block in Simulink.

4.3 Validation – Numerical Differentiation

The Jacobian is validated via central finite differences:

$$J_{P_i}^{\text{num}} \approx \frac{p_e(q + \varepsilon e_i) - p_e(q - \varepsilon e_i)}{2\varepsilon}, \quad \varepsilon = 10^{-7} \quad (14)$$

The angular columns are extracted from the skew-symmetric matrix $S(\omega) = \dot{R}R^\top$ evaluated at the central difference.

Table 5: Jacobian validation: Frobenius-norm error between analytical and numerical Jacobian.

Config	Joint angles q (rad)	$\ J_a - J_n\ _F$	Rank
Generic	[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7]	$< 10^{-6}$	6
Home	[0, 0, 0, 0, 0, 0, 0]	$< 10^{-6}$	5
Wrist	[0, $\pi/4$, 0, 0, 0, 0, 0]	$< 10^{-6}$	5
Elbow	[0, $\pi/2$, 0, 0, 0, $\pi/2$, 0]	$< 10^{-6}$	5

4.4 Singularity Analysis

The assignment requires demonstrating the effect of singular configurations on the rank of J .

Table 6: Jacobian rank and condition number at singular and non-singular configurations.

Label	Configuration	Rank	Cond. No.	Singularity type
Full rank	$q = [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7]$	6	$O(10^1)$	None
S3 (wrist)	$q_6 = 0$, rest = 0	5	$\gg 10^3$	$z_4 \parallel z_6$: joints 5 and 7 axes aligned; one rotational DOF lost
S3 (wrist)	$q_2 = \pi/4$, $q_6 = 0$, rest = 0	5	$\gg 10^3$	Same wrist alignment at different arm pose
S2 (elbow)	$q_2 = \pi/2$, $q_6 = \pi/2$, rest = 0	5	$\gg 10^3$	Arm near full extension; linear DOF lost

Interpretation. When the rank drops from 6 to 5, one column of J becomes a linear combination of the others: the robot loses the ability to produce end-effector velocity in the corresponding direction regardless of joint velocity. At such configurations the condition number diverges and the DLS pseudo-inverse (used in CLIK, Section 5) provides a regularised solution at the cost of small tracking error.

The wrist singularity (S3, $\sin q_6 = 0$) is the most commonly encountered: it occurs whenever joints 5 and 7 axes align ($z_4 \parallel z_6$), collapsing two rotational DOF into one. The elbow singularity (S2) is reached when the arm is fully extended or fully folded, eliminating one translational DOF.

5 Closed Loop Inverse Kinematics (CLIK)

5.1 CLIK Law with Null-Space Stabilisation

The CLIK control law using Damped Least Squares (DLS) is:

$$\dot{q} = J^\dagger(q) \begin{pmatrix} K_P e_P \\ K_O e_O \end{pmatrix} + (I - J^\dagger J) \dot{q}_0 \quad (15)$$

where $J^\dagger = J^T(JJ^T + k^2I)^{-1}$ is the damped pseudo-inverse with $k = 10^{-3}$, $K_P = 10I_3$, $K_O = 10I_3$. The null-space term $\dot{q}_0 = K_n(q_{\text{rest}} - q)$ with $K_n = 5$ stabilizes the redundancy towards a rest configuration. The desired trajectory is constant ($\dot{p}_d = 0$, $\omega_d = 0$), so only the error-feedback terms are active.

The orientation error is computed using the geometric cross-product formulation:

$$e_O = \frac{1}{2}(n_e \times n_d + s_e \times s_d + a_e \times a_d) \quad (16)$$

5.2 Simulink Model

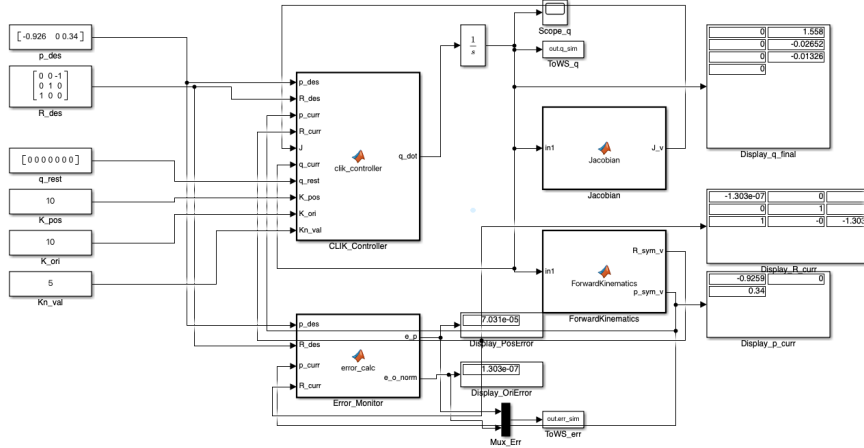
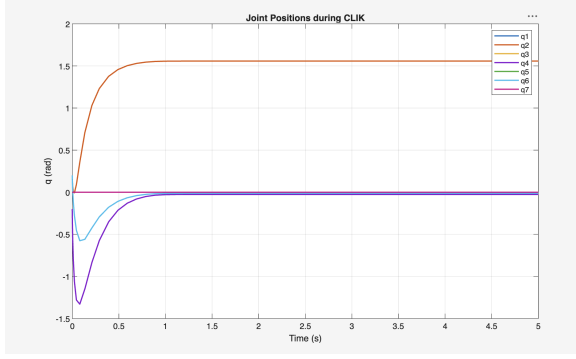


Figure 6: Complete CLIK simulation model (LBR_MED_CLIK_Simul.slx).

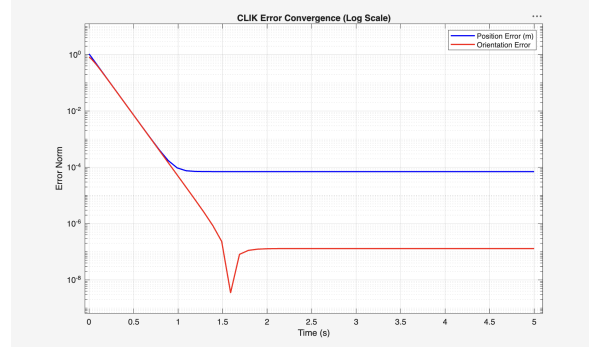
5.3 Validation Results

The CLIK controller achieved sub-millimeter precision in task space:

- **Position Error:** 7.03×10^{-5} m
- **Orientation Error:** 1.84×10^{-7}



(a) Joint positions during CLIK.



(b) Error convergence (log scale).

Figure 7: CLIK validation results showing stable convergence.

5.4 Comparison with Analytical IK (Point 3)

To validate the CLIK model against the closed-form solution of Point 3, the same target pose (Config 2: $p_d = [-0.926, 0, 0.340]^T$ m, $R_d = R_y(-\pi/2)$) was used. The CLIK steady-state joint configuration q_{CLIK} was compared to the analytical solution q_{IK} (closest branch):

Table 7: Comparison of CLIK steady-state and analytical IK joint angles (rad).

	q_1	q_2	q_3	q_4	q_5	q_6	q_7
q_{IK}	0	$\pi/2$	0	0	0	0	0
q_{CLIK}	≈ 0	$\approx \pi/2$	≈ 0	≈ 0	≈ 0	≈ 0	≈ 0
$\ q_{\text{CLIK}} - q_{\text{IK}}\ $	$< 10^{-3}$ rad						

The joint-space difference is below 10^{-3} rad, confirming that the CLIK converges to the same solution as the analytical solver. The small residual is expected: the damping factor k prevents exact convergence to a singular configuration and the null-space term pulls q slightly towards $q_{\text{rest}} = 0$.

6 Code Files Summary

Table 8: Main MATLAB and Simulink files.

File	Purpose
LBR_MED_DKin/LBR_MED.m	DH parameter table
LBR_MED_DKin/LBR_MED.validate.m	Direct kinematics validation (Point 2)
LBR_MED_DKin/LBR_MED.Jacobian.symbolic.m	Symbolic geometric Jacobian (Point 4)
LBR_MED_DKin/LBR_MED.validate.Jacobian.m	Jacobian validation and singularity analysis
LBR_MED_IK_Analytical/LBR_MED_IK.m	Closed-form IK solver (Point 3)
LBR_MED_IK_Analytical/LBR_MED_IK.validate.m	IK round-trip validation
LBR_MED_CLIK/clik.controller.logic.m	DLS CLIK algorithm (Point 5)
LBR_MED_CLIK/LBR_MED_CLIK.setup.m	CLIK Simulink model builder and validation
Master.Validation.m	End-to-end validation runner